

5	(a) amplitude between 6.5 squares and 7.5 squares on 3 peaks (allow 1 mark if outside this range but between 6.0 and 8.0 squares) correct phase (ignore lead/lag, look at x-axis only and allow $\pm\frac{1}{2}$ square)	B2	B1 [3]
	(b) $\lambda = ax / D$ $540 \times 10^{-9} = (0.700 \times 10^{-3} x) / 2.75$ $x = 2.12 \text{ mm}$	C1	C1 A1 [3]
	(c) (i) same separation bright areas brighter (1) dark areas, no change (1) (allow 'contrast greater' for 1 mark if dark/light areas not discussed) fewer fringes observed (1) any two, 1 each	B1	B2 [3]
	(ii) smaller separation of fringes no change in brightness	B1	B1 [2]
6	(a) power = VI	C1	